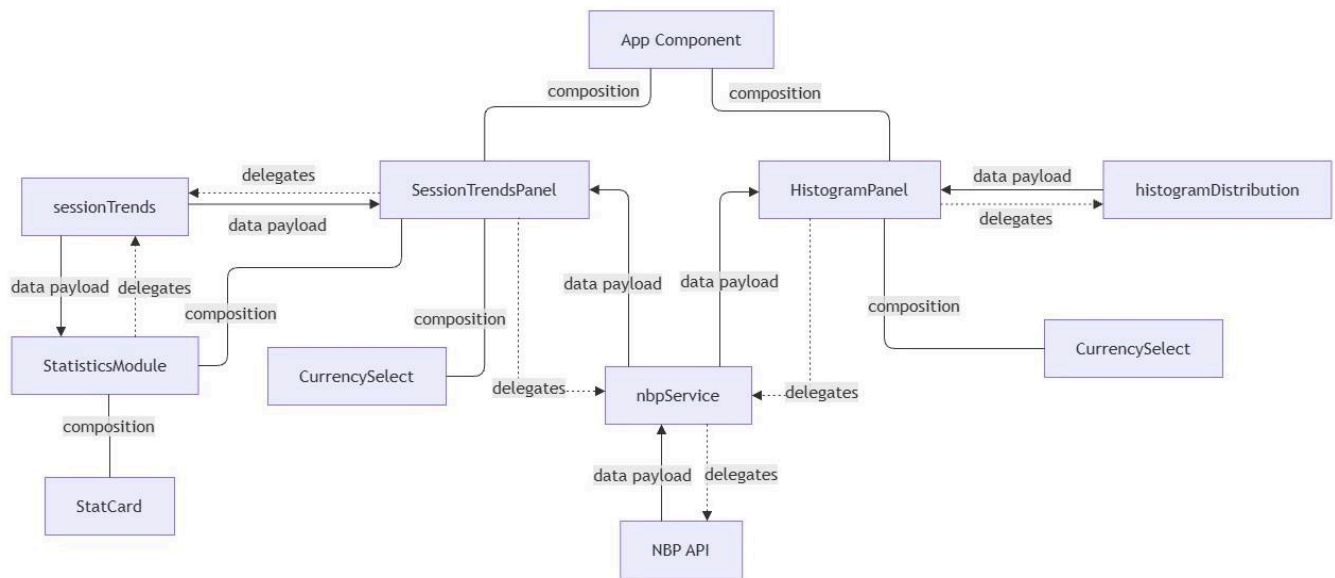


Component Diagram

The diagram models the structural architecture of the NBP Analytics application, illustrating the static dependencies and data flow between UI components, core utility modules, and the external NBP Web API. The structure is separated into three main layers: the entry/routing layer, the feature-specific panels layer, and the core business logic layer which handles web requests and calculations.



Application Initialization and Page Selection

The entry point of the application structure is the App Component. It acts as the orchestrator of the main layout, maintaining the active page state and rendering the appropriate view. The flow splits from the App Component through composition to instantiate either the SessionTrendsPanel or the HistogramPanel.

Session Trends Panel Workflow

The SessionTrendsPanel is a feature component that manages the workflow for analysis of currency exchange rate trends. It handles several direct dependencies:

- It uses the shared CurrencySelect component (via composition) to specify the selected currency.
- It delegates data fetching to the nbpService, receiving a data payload of raw exchange rates.
- It delegates to the sessionTrends module to perform calculations and receives a processed data payload.
- It instantiates the StatisticsModule via composition. The StatisticsModule itself connects to the sessionTrends module, delegating data to invoke core statistics functions and receiving a data payload, and it uses composition to render individual StatCard components.

Histogram Panel Workflow

The HistogramPanel manages the monthly and quarterly exchange rate change distribution workflow. Similar to the trends panel, it uses a local instance of the shared CurrencySelect component (via composition) to set the currency configurations. To handle API fetching, it delegates directly to nbpService and receives a data payload

of exchange rates. Once the rates are received, it delegates to histogramDistribution to process the data (such as calculating percentage changes and grouping them into bins), receiving the final processed data payload for visual rendering.

Core Services and External API Integration

The core business logic layer is separated into specialized modules to keep UI components focused on rendering:

- `nbpService`: Serves as the central data client for the application. It interacts directly with the external environment by delegating to the NBP API endpoint to retrieve exchange rate series. This acts as the final terminal link in the component graph. Both `SessionTrendsPanel` and `HistogramPanel` rely on this service for data.
- `sessionTrends` & `histogramDistribution`: These act as mathematical analysis utilities. `sessionTrends` is used by the trends panel and statistics module, while `histogramDistribution` is exclusively used by the histogram panel.